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Hematopoietic stem cell transplantation disrupts age-related gut microbiota signatures in pediatric and adult recipients

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Allogeneic hematopoietic stem cell transplantation (allo-HSCT) profoundly reshapes the gut microbiome (GM), yet age-related differences in this process remain incompletely understood. In this study, we analyzed stool samples from a cohort of pediatric and adult patients collected before allo-HSCT and at neutrophil engraftment to assess GM dynamics. We observed a significant reduction in alpha diversity post-HSCT across both age groups, with young patients displaying notably lower biodiversity. While GM composition varies among age ranges before transplantation, these distinctions largely disappeared at neutrophil engraftment, indicating a convergence toward a more uniform “transplant-specific” signature. This shift entailed a loss of health-associated, butyrate-producing taxa and a rise in potentially opportunistic bacteria (e.g., *Enterococcaceae*, *Staphylococcaceae*). Low pre-allo-HSCT GM diversity correlated with an increased risk of clinically significant acute graft-versus-host disease, overwhelming the age differences, underscoring the clinical relevance of maintaining a balanced microbiome before transplantation. Overall, our findings highlight the importance of age in shaping the pre-transplant GM and reveal how allo-HSCT-driven factors homogenize microbial communities among age ranges, offering insights for future microbiome-targeted interventions to improve transplant outcomes.

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INTRODUCTION

Specific configurations of the gut microbiome (GM) have been associated with clinical outcomes in both adult and pediatric allogeneic hematopoietic stem cell transplantation (allo-HSCT) recipients [1–6]. In adults, higher alpha diversity prior to transplant and at neutrophil engraftment has been associated with a higher probability of overall survival (OS) and lower incidence of graft-versus-host disease (GvHD)-related mortality in T-replete grafts [2]. Moreover, several GM signatures in the pre- and post-transplant period have been correlated with increased risk of GvHD, particularly the overgrowth of *Enterococcus* [7], the loss of commensals, such as *Blautia* [8], and the reduced production of short-chain fatty acids (SCFAs) [9]. GM signatures before HSCT have also been shown to predict GvHD, infection complications, and survival in children, including lower alpha diversity [1], higher abundance of arginine-producing bacteria [10], lower *Blautia* content, and an increase in *Fusobacterium* abundance [4]. However, pediatric studies show more variable associations between post-allo-HSCT GM modifications and outcomes [11]. While shifts in GM diversity following HSCT are well-documented in both adults and children [12, 13], it remains unclear whether these alterations are influenced by the recipient's age and how age impacts the connection between GM and outcomes. Notably, evidence from adult cohorts is supported by significantly larger

sample sizes, which enhances data reliability and reduces susceptibility to biases. On the contrary, pediatric studies reported smaller cohorts and greater variability, which represent a pronounced limitation. Age-related differences in GM composition are well-documented in healthy individuals, with the GM changing its compositional and functional profile from infancy to old age, following a developmental trajectory across different life stages [14]. For instance, younger populations typically harbor higher abundances of beneficial taxa such as *Bifidobacteriaceae* [15], while older individuals exhibit increased representation of Proteobacteria and opportunistic pathogens [16]. Moreover, the pediatric GM undergoes larger shifts when exposed to external stressors compared to adults [15]. However, whether these pre-existing microbial distinctions persist post-HSCT or converge toward a common post-transplant profile remains an open question [12]. Furthermore, differences in transplant procedures between adults and pediatric patients may affect the GM in specific ways. For example, myeloablative conditioning is more frequently administered in pediatrics, causing greater microbiome disruption compared to reduced-intensity regimens [17]. Moreover, the incidence of key clinical outcomes examined in microbiome studies, such as GvHD, differs significantly between adults and children, further complicating direct comparisons and the interpretation of GM-related findings. To address this gap, we

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analyzed the GM of a mixed cohort of children and adult patients undergoing allo-HSCT at the adult and pediatric transplant center of the same hospital. By investigating changes in microbial diversity and taxonomic composition across different age groups, we aim to determine whether and how age-related GM signatures influence post-HSCT microbial dynamics.

METHODS

Study design and definitions

We conducted a prospective study enrolling patients undergoing allo-HSCT at the pediatric and adult transplant units at the IRCCS Azienda Ospedaliero-Universitaria di Bologna. In the hospital, the antibiotic stewardship program is shared between the adult and pediatric transplant units. The study was approved by the Ethics Committee of the enrolling institution (reference number 19/2013/U/Tess) and was conducted in accordance with the Declaration of Helsinki. All patients, or their parents/legal guardians, provided written informed consent to participate in the study. Patients receiving allo-HSCT were enrolled in the study from January 2014 to October 2023. The exclusion criteria included having received fecal microbiota transplantation or pre/probiotics. Patients from the pediatric unit were previously published as part of a larger cohort [1]. A stool sample was collected from each patient before the start of the conditioning regimen and at the time of neutrophil engraftment. Neutrophil engraftment was defined as the first of three consecutive days in which the neutrophil count was greater than $0.5 \times 10^9/L$. GvHD was graded according to the Mount Sinai Acute Gvhd International Consortium (MAGIC) criteria [18]. Clinical outcomes included the cumulative incidence of acute GvHD (aGvHD) and the graft-versus-host disease-free, relapse-free survival (GRFS), according to the GM configuration. Antibiotic prophylaxis was administered according to the center's policy and consisted of fluoroquinolone prophylaxis, which started at the time of the conditioning regimen and was discontinued at the onset of febrile neutropenia or upon neutrophil engraftment.

Microbiome analysis

After collection, stool samples were stored at 4 °C for a maximum of 48 h before being frozen at −80 °C. Methods for DNA extraction, 16S ribosomal RNA (rRNA) amplicon sequencing, and data processing for the cohort were applied as previously reported [4, 19, 20]. A comparable methodology was applied to the adult cohort. In brief, stool samples were processed to extract and purify microbial DNA using a modified bead-beating method [20], involving lysis, nucleic acid precipitation, and removal of contaminants. DNA purification was achieved with the DNeasy Blood and Tissue Kit (QIAGEN, Hilden, Germany) according to the manufacturer's instructions. DNA concentration and quality were evaluated using a Nanodrop ND-1000 spectrophotometer (NanoDrop Technologies, Wilmington, DE, USA). Libraries were prepared targeting the V3–V4 region of the 16S rRNA gene using the 341F and 785R primers. Amplification was performed using the KAPA HiFi HotStart ReadyMix (Roche, Basel, Switzerland), while indexing was carried out with Illumina Nextera technology. Clean-up steps were completed using a magnetic bead-based system (Agencourt AMPure XP; Beckman Coulter, Brea, CA, USA). Sequencing was performed on an Illumina MiSeq platform using a 2 × 250 bp paired-end protocol, as per the manufacturer's guidelines. Raw sequences were processed with the publicly available containerized pipeline 16S-KrakenBracken-Pipeline [21]. This pipeline integrates quality filtering with fastp [22], read mapping and taxonomy assignment against the SILVA 138_1 SSURef database using Kraken2 [23] and Bracken [24], and diversity calculation performed with the vegan [25], otuSummary [26], and phyloseq [27] R packages. New sequencing data produced were deposited in the National Center for Biotechnology Information Sequence Read Archive (NCBI SRA, PRJNA 1263152). Previously obtained sequence samples are publicly available as previously detailed [1].

Statistical analysis

Clinical data were collected via electronic case report forms. Data were reported using descriptive statistics. Continuous variables were presented as median values, range, or standard deviation (SD). The variables were compared between patients developing grades II–IV aGvHD and those who did not, using the Wilcoxon rank sum test for continuous variables and the chi-square (χ^2) test for categorical variables. The incidence of aGvHD was estimated using the cumulative incidence function to account

for competing risks. Grey's test was used to compare the data between the two groups. GRFS were estimated using the Kaplan–Meier method and Log-rank, and were used to assess statistical differences. Multivariate analysis was performed using a Cox regression model to evaluate the risk of aGvHD, refined for the competitive events, namely, death and graft failure. A *p* value of <0.05 was considered statistically significant. Clinical statistical analyses were performed using NCSS (NCSS 12 Statistical Software [2018]). NCSS, LLC, Kaysville, UT. Microbiological statistical analyses were performed using R software (v4.4.2) [28]. The relative abundances of phyla, families, and genera over time were graphically represented using boxplots created with the microbAIDeR R package [29]. Principal Coordinate Analysis (PCoA) plots were generated with the “vegan” package (version 2.6-2, Oksanen et al. [25]), and data separation was tested using a permutation test with pseudo-F ratio (function “Adonis” in “vegan”). Differences in alpha diversity and relative taxon abundances at phylum, family, and genus levels among groups were assessed using Kruskal–Wallis tests followed by post-hoc Wilcoxon tests. Family- and genus-level heatmaps were constructed with the “made4” [30] package and the “heatmap.2” function from the “gplots” package [31]. To explore intrinsic patterns of GM variation among adult and pediatric subjects undergoing HSCT, a clustering analysis was performed. Samples were grouped into clusters based on their genus-level GM composition using k-means clustering. Cluster quality was evaluated with the “factextra” R package using the silhouette approach [32]. Genera characterizing each cluster were identified with the random forest-based feature selection algorithm Boruta, implemented via the “Boruta” R package [33], with the following parameters: *p*Value = 0.01, *mcAdj* = TRUE, *maxRuns* = 10,000. Analyses were stratified by age group (≥ 15 years or <15 years). The impact of antibiotic prophylaxis on gut microbiota composition was assessed using MaAsLin2 [34], including antibiotic prophylaxis as a fixed effect alongside age and timepoint, with TSS normalization and log transformation. *P* values were corrected for multiple comparisons using the Benjamini–Hochberg method, where appropriate. A false discovery rate ≤ 0.05 was considered statistically significant, while a *p* value between 0.05 and 0.1 was considered indicative of a trend.

RESULTS

Patient's characteristics

A total of 88 patients were enrolled in the study and stratified into two groups to account for age-related differences, based on if patients were younger than 15 years (36 patients) or older (52 patients). Patient characteristics are summarized in Table 1. The median age at allo-HSCT was 7.2 years (range, 0.9–14.4) in the younger cohort and 45.1 years (range, 15.4–70.2) in the older cohort. Males accounted for 36.9% of patients under 15 years and 63.0% of those over 15 years. The most common indications for allo-HSCT in the overall cohort were acute myeloid leukemia (36.4%), acute lymphoblastic leukemia (26.1%), and myelodysplastic syndrome/myeloproliferative neoplasm (13.6%). The disease distribution varied by age group, with acute lymphoblastic leukemia being the predominant diagnosis in younger patients (47.2%), while acute myeloid leukemia was more frequent in older patients (46.2%). Nonmalignant diseases accounted for 12.5% of cases, with a higher prevalence in the younger cohort (16.7%) compared to the older cohort (9.6%). Regarding stem cell source, bone marrow (BM) was predominantly used in younger patients (97.2%), whereas peripheral blood stem cells (PBSC) were the primary source in the older group (75.0%). Most patients in both groups underwent transplantation from unrelated donors (63.9% in the younger cohort and 82.7% in the older cohort). Mucositis was more frequently observed in younger patients, occurring in 45 patients (55%), with 11 cases (24%) of grade ≥ 3 , compared to 13 patients (15%) in the older group, of whom 5 (38%) had grade ≥ 3 mucositis (*p* < 0.001). The median duration of mucositis was 12 days (range, 12–35) in the younger group and 10 days (range, 6–14) in the older group, with no statistically significant difference (*p* = 0.291). Myeloablative conditioning was more common in patients under 15 years (66.7%), while reduced-intensity regimens were more frequently used in older patients (55.8%). Antibiotic prophylaxis was administered in 50% of patients overall, though it

Table 1. Patients' characteristics.

Characteristic	Overall (%)	<15 years (%)	>15 years (%)
No. of patients	88	36	52
Age at allo-HSCT, years (range)	29.6 (0.9–70.2)	7.2 (0.9–14.4)	45.1 (15.4–70.2)
Male sex, <i>n</i> (%)	46 (52.2)	17 (36.9)	29 (63.0)
Disease, <i>n</i> (%)			
Acute myeloid leukemia	32 (36.4)	8 (22.2)	24 (46.2)
Acute lymphoblastic leukemia	23 (26.1)	17 (47.2)	6 (11.5)
Myelodysplastic syndrome or myeloproliferative neoplasm	12 (13.6)	3 (8.3)	9 (17.3)
Non-Hodgkin's lymphoma	5 (5.7)	0 (0)	5 (9.6)
Chronic myeloid leukemia	3 (3.4)	0 (0)	3 (5.8)
Juvenile Myelomonocytic Leukemia	2 (2.3)	2 (5.6)	0 (0)
Nonmalignant disease	11 (12.5)	6 (16.7)	5 (9.6)
Stem cell source (%)			
unmodified BM	48 (55.4)	35 (97.2)	13 (25.0)
unmodified PBSC	40 (45.4)	1 (2.18)	39 (75.0)
Type of donor (%)			
Unrelated	66 (75.0)	23 (63.9)	43 (82.7)
Identical sibling	12 (13.6)	6 (16.7)	6 (11.5)
HLA-haploidentical sibling	10 (11.4)	7 (19.4)	3 (5.7)
Intensity of conditioning (%)			
Myeloablative	47 (53.4)	24 (66.7)	23 (44.2)
Reduced intensity	41 (46.6)	12 (33.3)	29 (55.8)
Antibiotic prophylaxis (%)			
No	44 (50)	26 (72.2)	18 (34.6)
Yes	44 (50)	10 (27.8)	34 (65.4)

HSCT hematopoietic stem cell transplantation.

was more commonly used in the older cohort (65.4%) than in the younger cohort (27.8%). From the enrolled patients, a total of 176 stool samples were analyzed in this study. On average, 93131.43 ± 76348.85 SD high-quality 16S rRNA gene sequences were obtained per sample.

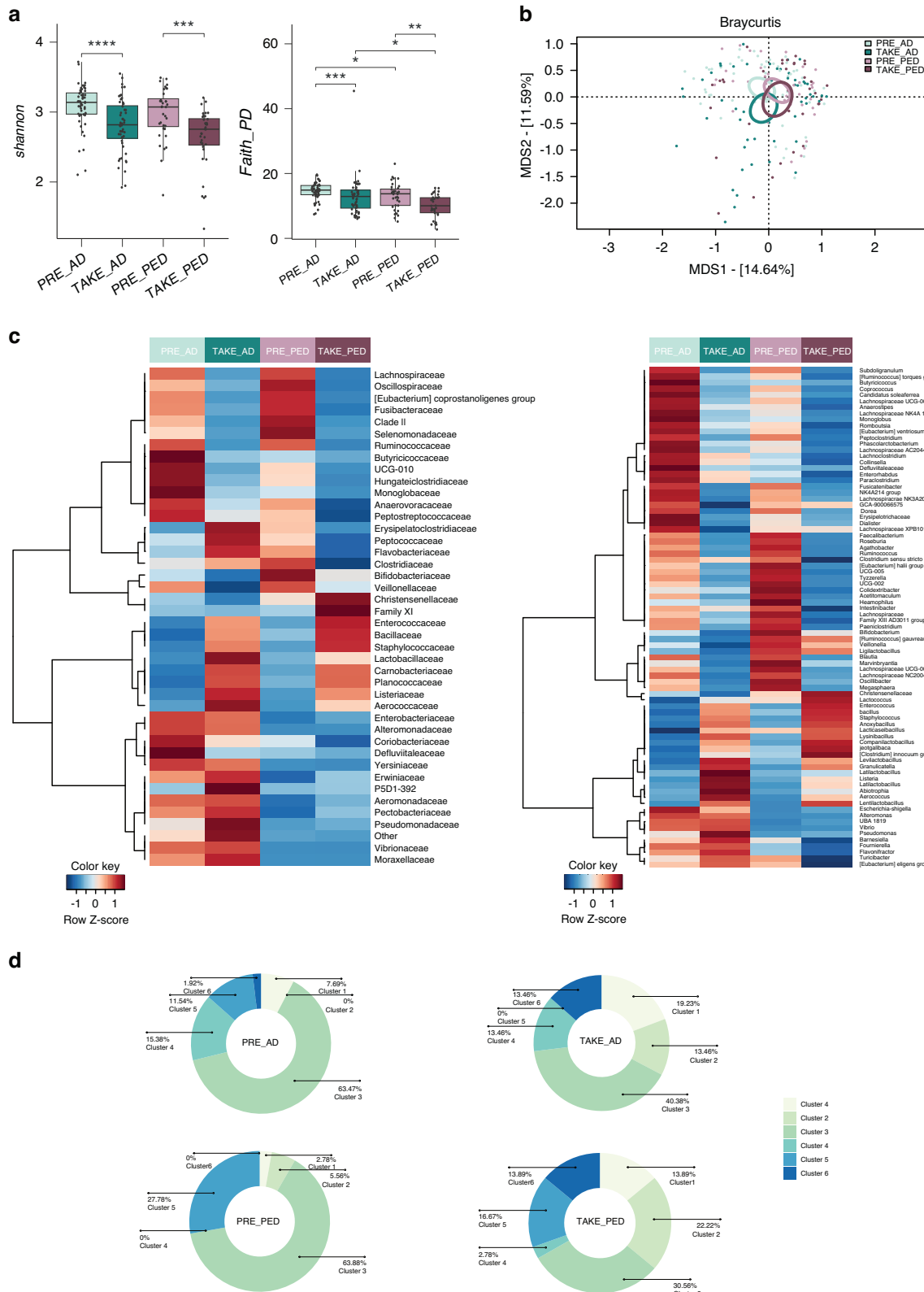
Allo-HSCT alters GM diversity similarly among pediatric and adult recipients

Alpha diversity was assessed using two metrics: Faith's Phylogenetic Diversity (Faith's PD) and Shannon entropy. Both metrics revealed a statistically significant decrease in biodiversity following HSCT in both patient groups (Wilcoxon test, $p \leq 0.001$). Patients under 15 years exhibited lower biodiversity than older ones at both time points when measured with Faith's PD ($p < 0.02$) (Fig. 1a). Beta diversity analysis, based on Bray–Curtis distances, showed a significant separation in GM composition between pre-transplant and neutrophils engraftment time points in both groups (PERMANOVA, $p \leq 0.001$) (Fig. 1b). Additionally, the GM of older cohort differed significantly from that of the younger cohort before the transplant ($p = 0.04$); however, this difference was no longer observed after transplant ($p = 0.20$). Additionally, GM diversity significantly correlated with age both before and after HSCT based on Faith's PD ($p \leq 0.02$), whereas with Shannon entropy, a significant correlation was observed only before HSCT ($p = 0.02$) (Supplementary Fig. 1). We also analyzed Bray–Curtis dissimilarity prior to HSCT in patients with acute leukemia compared to others and did not find any significant differences.

Compositional shifts occur after HSCT, depleting the age-related signature

We then analyzed the compositional differences between the two groups at both time points. Differences at the phylum level were

observed in pre-transplant, as detailed in the Supplementary Text 1. GM profiles at engraftment revealed notable similarities between younger and older patients' family level composition, with consistent trends observed across multiple taxa (Fig. 1c, Supplementary Text 1, Supplementary Fig. 2). At the genus level, the most abundant genera characterizing the GM of both age groups were *Bacteroides* ($9.8\% \pm 16\%$), *Streptococcus* ($8.7\% \pm 12.7\%$), *Bifidobacterium* ($6.9\% \pm 11.7\%$), *Faecalibacterium* ($5.4\% \pm 7.2\%$), and *Escherichia-Shigella* ($5.2\% \pm 10.8\%$). Pre-HSCT differences were evident between age groups. Older patients exhibited increased abundances of *Butyrivibrio*, *[Ruminococcus] torques* group, *Anaerostipes*, *Lachnospiraceae* NK4A136 group, *Escherichia-Shigella*, *Collinsella*, *Pantoea*, *Serratia*, *Enterorhabdus*, *Pseudomonas*, and *Lactobacillus* ($p < 0.03$). In contrast, younger patients showed higher levels of *Bifidobacterium* and *Lachnospiraceae* UCG-008 ($p < 0.05$). Among these genera, only *Butyrivibrio*, *Anaerostipes*, *Escherichia-Shigella*, *Pantoea*, *Serratia*, *Enterorhabdus*, and *Pseudomonas* displayed trends toward differential abundance between the two groups at the neutrophil engraftment stage ($p \leq 0.09$). In fact, taxonomic shifts following HSCT paralleled those observed at the family level, with consistent trends within the corresponding genera (Fig. 1c, Supplementary Text 1, Supplementary Fig. 3). These findings underscore the disruption of age-dependent microbial patterns, with pre-HSCT distinctions between younger and older patients' GM profiles largely disappearing post-HSCT, except for a few microbial signatures retaining differential trends. Furthermore, to rule out potential confounding factors, the impact of antibiotic prophylaxis on GM composition was explored using multivariable analysis, where we included antibiotic prophylaxis as a fixed effect alongside age and timepoint. We observed that while antibiotic prophylaxis was



associated with minor changes in alpha diversity and the relative abundance of specific genera, compared to the strongest effects of age and the HSCT procedure itself. Notably, the reduction in GM diversity after HSCT was evident even in pediatric patients who

did not receive antibiotics, suggesting that the transplant process is a key driver of these changes (Supplementary Text 2, Supplementary Fig. 4).

Fig. 1 HSCT alters gut microbiota composition and leads to loss of age-related features. Boxplots showing the distribution of alpha diversity, estimated with the Shannon index (left) and Faith's Phylogenetic Diversity (Faith_PD; right), for samples taken from patients greater than/equal to 15 years before HSCT ("PRE_AD") or at neutrophils engraftment ("TAKE_AD") and from patients less than 15 years of age before HSCT ("PRE_PED") or at neutrophils engraftment ("TAKE_PED"). Wilcoxon test, * p value ≤ 0.05 ; ** p value ≤ 0.01 ; *** p value ≤ 0.001 ; **** p value ≤ 0.0001 (a). Principal Coordinates Analysis (PCoA) based on Bray–Curtis distances between "PRE_AD," "TAKE_AD," "PRE_PED," and "TAKE_PED" samples. A significant separation was observed between samples taken before and after HSCT in both groups of patients and "PRE_AD" and "PRE_PED" samples (permutation test with pseudo-F ratios (Adonis), $p < 0.04$) (b). Heatmap of the mean relative abundance of taxa at the family level (left) and genus level (right). Only taxa with at least one comparison showing a significant p value (Wilcoxon test) are displayed (c). Pie chart showing the distribution of "PRE_AD," "TAKE_AD," "PRE_PED," and "TAKE_PED" samples in the six clusters obtained by applying K-means clustering (d).

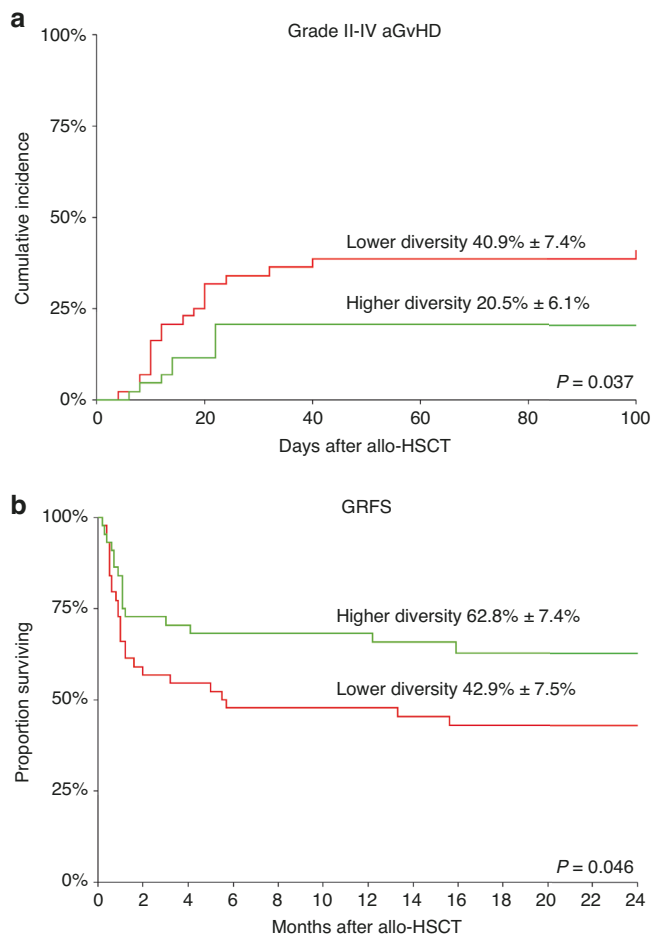


Fig. 2 Cumulative incidence of grades II–IV aGvHD and GRFS. **a** Cumulative incidence according to pre-allo-HSCT GM diversity for grades II–IV aGvHD; **b** Kaplan–Meier curves of GRFS stratified by pre-allo-HSCT GM diversity.

Clustering highlights the loss of age-related GM signatures

To investigate intrinsic patterns of variation in GM structure among younger and older subjects undergoing allo-HSCT, k-means unsupervised clustering was performed based on the relative abundances of the microbial components at the genus level, regardless of age groups and transplant timepoint, identifying six distinct clusters (Clusters 1–6; Supplementary Fig. 5, Fig. 1D). Feature selection allowed us to identify 31 taxa as the most important contributors to cluster differentiation (Supplementary Fig. 6). Cluster 1 was predominantly composed of *Bacteroides*, *Prevotella*, *Prevotella_9*, and *Porphyromonas*. Cluster 2 featured *Enterococcus*, *Bacillus*, *Lactococcus*, *Lysinibacillus*, *Granulicatella*, *Staphylococcus*, *Lentilactobacillus*, and *Levilactobacillus*. Cluster 3 did not exhibit any distinct microbial signature but

showed slightly higher abundances of *Butyrivibrio*. Cluster 4 was enriched with *Vibrio*, *Salmonella*, *Serratia*, *Escherichia-Shigella*, *Pantoea*, and *Haemophilus*. Cluster 5 was primarily identified by *Bifidobacterium*, while Cluster 6 was dominated by *Streptococcus*. Pre-HSCT, older patients' samples were predominantly distributed across Clusters 3 (63.5%), 4 (15.4%), 5 (11.5%), and 1 (7.7%) (Fig. 1D). In contrast, samples from younger patients were largely confined to Clusters 3 (64%) and 5 (27.8%). This age-dependent clustering pattern highlights distinct microbial signatures between the two groups before transplantation. Following HSCT, the dominance of Cluster 3 diminishes in both groups, with a more balanced distribution across clusters, indicating partial convergence of microbial profiles. Specifically, post-HSCT samples were distributed across Clusters 1 (older: 19.2%; younger: 14%), 2 (older: 13.5%; younger: 22.2%), and 6 (older: 13.9%; younger: 13.5%). Notably, a significant proportion of samples from older patients remained in Cluster 4 (13.5%), while samples from younger ones retained a presence in Cluster 5 (16.7%).

GM diversity is associated with graft-versus-host disease regardless of age

To assess the relationship between GM composition in younger and older patients developing aGvHD, we first divided the patients into two groups based on GM diversity before allo-HSCT, as previously reported [1, 2]. Patients with higher GM diversity before allo-HSCT exhibited a significantly lower cumulative incidence of 100-day grades II–IV aGvHD (20.5 ± 6.1 vs 40.9 ± 7.4 , $p = 0.037$) and a significantly higher 2-year GRFS (62.8 ± 7.4 vs 42.9 ± 7.5 , $p = 0.046$) (Fig. 2). To address differences between patients with and without grades II–IV aGvHD, we compared the baseline characteristics (Supplementary Table 1) and found no significant differences. We also performed a multiparametric Cox regression model accounting for transplant characteristics and age group and found that only the use of anti-thymocyte globulins and pre-transplant GM diversity were significantly associated with grades II–IV aGvHD (Table 2). We also analyzed potential pre-transplant factors associated with low diversity, including underlying disease, age at transplant, and antibiotic exposure prior to conditioning, but no significant associations were found. No differences were found in OS ($p = 0.525$), event-free-survival ($p = 0.640$), and with diversity at neutrophil engraftment.

GM age-related features in aGvHD patients

To further explore the association between low GM biodiversity and the increased incidence of clinically significant aGvHD, we analyzed GM dynamics at the genus level across patients developing aGvHD. Patients were stratified into two groups: those with no or low-grade aGvHD (aGvHD[0–1]) and those with clinically significant aGvHD (aGvHD[2–4]). Prior to transplantation, patients who later developed clinically significant aGvHD had significantly lower relative abundances of the *Eubacterium hallii* group and the *Lachnospiraceae* NC2004 group compared to those in the aGvHD[0–1] group ($p \leq 0.04$) (Fig. 3). At neutrophil engraftment, the aGvHD[0–1] group showed higher abundances of *Bacillus*, *Lactobacillus*, *Granulicatella*, *[Ruminococcus] gausvreauii* group, *Levilactobacillus*, *Limosilactobacillus*, *Lactiplantibacillus*,

Table 2. Multivariate analysis of grades II–IV aGvHD according to transplant characteristics.

Independent variable	Odds ratio (OR) $\pm$ SE	p value
Pre-HSCT diversity below the median value vs above	5.08 $\pm$ 1.13	0.015
Age group >15 years vs <15 years	2.06 $\pm$ 0.72	0.360
Male vs female sex	0.77 $\pm$ 0.54	0.633
Oncological disease vs non-oncological	0.68 $\pm$ 0.81	0.649
MFD donor vs other	3.48 $\pm$ 1.09	0.253
MUD donor vs others	2.95 $\pm$ 1.29	0.401
PBSC vs BM	3.69 $\pm$ 0.67	0.051
Reduced intensity conditioning vs myeloablative	0.58 $\pm$ 0.54	0.331
ATG vs no ATG	0.08 $\pm$ 1.11	0.027

In bold, the *p* values < 0.05 are considered statistically significant.

Holdemanella, and *Enterococcus* ($p \leq 0.05$). Additionally, *Pediococcus* and *Lactococcus* displayed a trend of increase in this group post-HSCT ($p \leq 0.07$). In contrast, patients with clinically significant aGvHD exhibited a trend toward an increased abundance of the *Ruminococcus gnavus* group at the neutrophil engraftment stage ($p = 0.07$).

We further refined our analysis by examining GM composition before HSCT in both younger and older individuals, stratifying them based on the presence or absence of aGvHD (Supplementary Fig. 7). Regardless of GvHD status, older patients exhibited higher proportions of *Enterobacteriaceae* (including *Escherichia-Shigella*), *Vibrionaceae* (including *Vibrio*), *Pseudomonadaceae* (including *Pseudomonas*), and *Erwiniaceae* than younger ones ($p \leq 0.09$). Additionally, patients >15 years without aGvHD had significantly higher abundances of *Butyricicoccaceae* (and *Butyricicoccus*), *Staphylococcaceae* (and *Staphylococcus*), *Muribaculaceae*, *Lactobacillus*, and *Pediococcus* compared to the younger population without GvHD ($p < 0.05$). Among older patients without aGvHD, *Bacillaceae* and *Bacillus* are more abundant than in both adults with aGvHD and younger individuals without aGvHD ($p < 0.03$), while *Carnobacteriaceae* are also elevated relative to older patients with aGvHD ($p = 0.005$). In contrast, *Bifidobacteriaceae* (including *Bifidobacterium*) and *Clostridium sensu stricto* 1 are more abundant in children without aGvHD compared to adults without aGvHD ($p \leq 0.02$). Younger individuals without aGvHD display a higher relative abundance of *Peptostreptococcaceae*, *Lachnospiraceae* UCG-008, *Peptoclostridium*, *Intestinibacter*, and *Romboutsia* than patients without aGvHD ($p < 0.03$), although *Romboutsia* levels are increased in older individuals with aGvHD compared to those without ($p = 0.03$). Furthermore, younger individuals with aGvHD exhibit a reduced relative abundance of *Anaerovoracaceae*, *Anaerostipes*, [*Ruminococcus*] *torques* group, *Lachnospiraceae* NK4A136 group, and *Ruminococcus* compared to adults with aGvHD and children without GvHD ($p < 0.03$).

Microbiota features across age subgroups

To better account for the substantial influence of age on GM composition, we stratified the cohort into four distinct age groups: 0–3 years (“infants,” 10 patients), 4–14 years (“children,” 26 patients), 15–39 years (adolescents and young adults, or “AYA,” 17 patients), and 40–70 years (“adults,” 35 patients). This refined age-based stratification allowed for a more granular analysis of GM dynamics, minimizing age-related variation as a confounding factor and enabling a deeper understanding of GM changes across different life stages during HSCT. We specifically focused on

observing the dynamics of those taxa identified in previous studies by Peled et al. [2] and Masetti et al. [1] as key indicators of gut ecosystem characteristics in adults and pediatric patients undergoing HSCT. Figure 4 illustrates the trajectories of key taxa across the stratified age groups, revealing distinct age- and transplant-dependent patterns. *Bifidobacterium* and *Lactococcus* displayed age-related patterns. Pre-HSCT, *Bifidobacterium* levels were significantly higher in infants compared to adults, with children showing higher levels than both AYAs and adults ($p \leq 0.08$). Similarly, *Lactococcus* was most abundant in infants compared to children and adults pre-HSCT, but a significant post-HSCT increase was observed only in children and adults ($p \leq 0.008$). *Veillonella* was elevated in infants compared to children pre-HSCT, with all groups showing a decline post-HSCT ($p \leq 0.06$). Among *Enterobacteriaceae*, *Escherichia-Shigella* and *Enterobacter* display marked distinct age- and transplant-dependent variations. Pre-HSCT, both genera were more abundant in infants and adults than in children and AYAs ($p \leq 0.02$). Post-HSCT, *Enterobacter* declined in infants and adults but increased in AYAs ($p \leq 0.04$). *Enterococcus* and *Streptococcus* were most abundant in infants pre-HSCT, with *Enterococcus* increasing significantly post-HSCT in all groups except infants, where the change was not statistically significant ($p \leq 0.09$). *Collinsella* was most abundant in adults pre-HSCT, with post-HSCT declines observed in all groups except infants ($p \leq 0.06$). *Bacteroides* consistently showed low levels in infants both pre- and post-HSCT ($p < 0.02$). Several butyrate-producing taxa, including *Ruminococcus*, *Faecalibacterium*, and *Subdoligranulum*, were generally lower in infants compared to older groups pre-HSCT ($p \leq 0.07$). Notably, children exhibited higher levels of *Faecalibacterium* than adults ($p = 0.04$, respectively). These taxa showed significant declines post-HSCT across all age groups ($p \leq 0.04$). The [*Ruminococcus*] *gnavus* group was more abundant in infants compared to children and adults pre-HSCT and declined post-HSCT only in infants and adults ($p \leq 0.02$). *Lachnospiraceae* levels dropped across all groups post-HSCT ($p \leq 0.05$), with AYAs showing higher levels than adults at both time points ($p < 0.03$). *Blautia* followed a similar trend, with pre-HSCT levels higher in AYAs compared to adults and infants, though the change in infants post-HSCT was not significant ($p \leq 0.08$). Finally, *Staphylococcus* increased post-HSCT across all groups, with pre-HSCT levels higher in children than in infants and adults ($p < 0.03$).

DISCUSSION

Microbiome studies have provided significant insights in recent years into its role in the setting of allo-HSCT. However, studies in this setting were strongly divided in pediatric [1, 10, 20, 35, 36] and adult ones [2, 7–9], with both overlapping and diverging results. Indeed, as recently pointed out [11], the differences between pediatric and adult studies may be explained by the greater dynamism of the intestinal microbiome in childhood, which continues evolving until later in childhood. We report for the first time a microbiome study including both pediatric and adult patients from the same institution, ensuring that all patients were exposed to a shared hospital environment. This reduces the impact of the exposome and helps mitigate potential biases related to hospital-specific microbial colonization. Our data confirms significant and profound shifts in the GM composition of both children and adults undergoing allo-HSCT. Among the most prominent findings was a decrease in both alpha diversity after transplantation, as largely documented [12]. In fact, this observation aligns with previous reports, likely due to the cumulative effects of drugs such as chemotherapy, antibiotics, proton-pump inhibitors, the changes in nutritional intake, and the allo-immunity [2, 8, 12, 19, 37, 38]. In our cohort, younger patients demonstrated particularly lower biodiversity, which may reflect

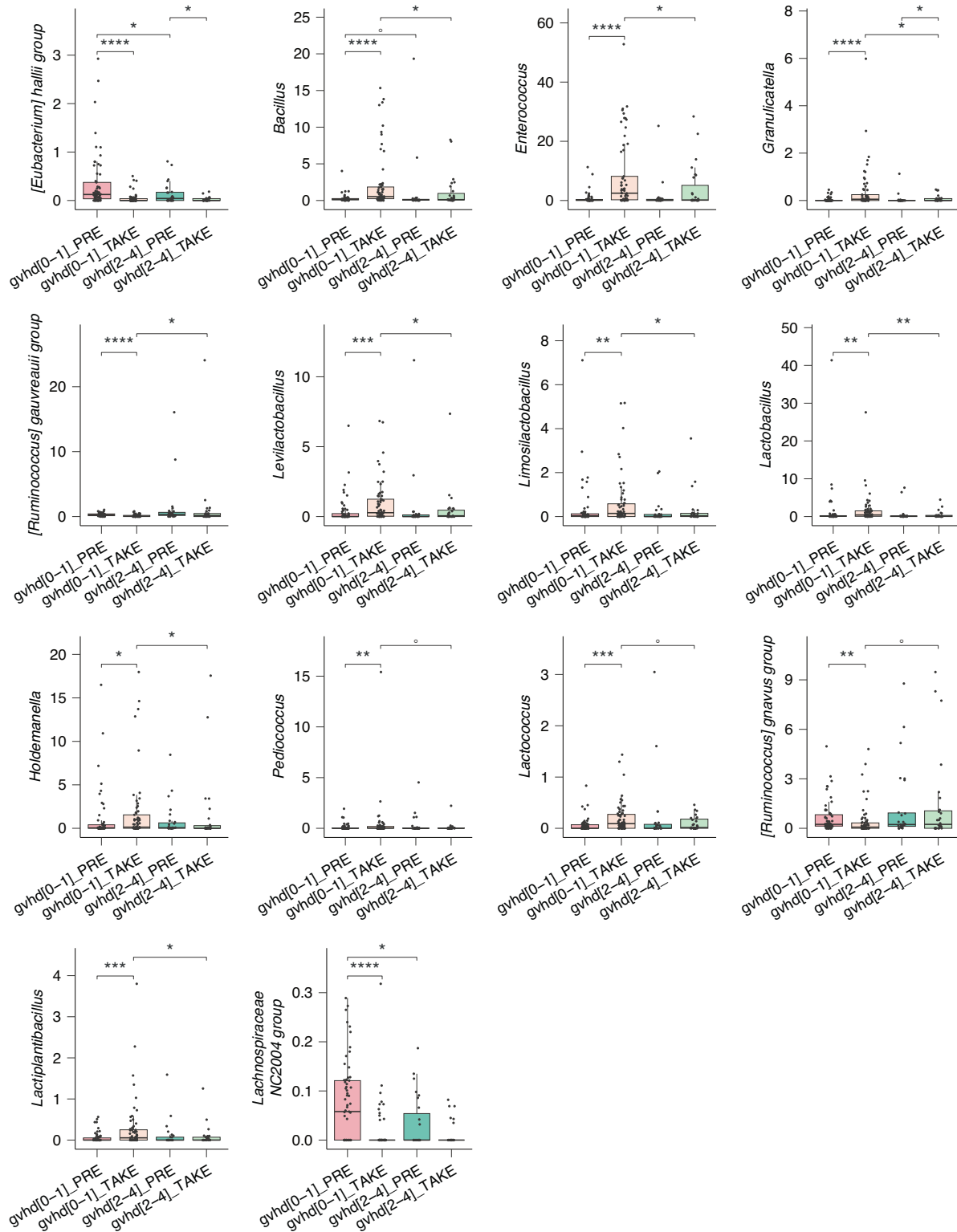


Fig. 3 Gut microbiota genera are differentially represented before HSCT and at neutrophil engraftment in patients who developed no or low-grade aGvHD and clinically significant aGvHD. Boxplots showing the relative abundance (%) distribution of genera of the gut microbiota differentially represented between samples from patients who developed no or grade 1 aGvHD before HSCT ("GvHD[0-1]_PRE") and at neutrophils engraftment ("GvHD[0-1]_TAKE") and those from patients with grade 2-4 aGvHD before HSCT ("GvHD[2-4]_PRE") and at neutrophils engraftment ("GvHD[2-4]_TAKE"). Wilcoxon test, * p value ≤ 0.05 ; ** p value ≤ 0.01 ; *** p value ≤ 0.001 ; **** p value ≤ 0.0001 ; $^{\circ}$ p value ≤ 0.09 .

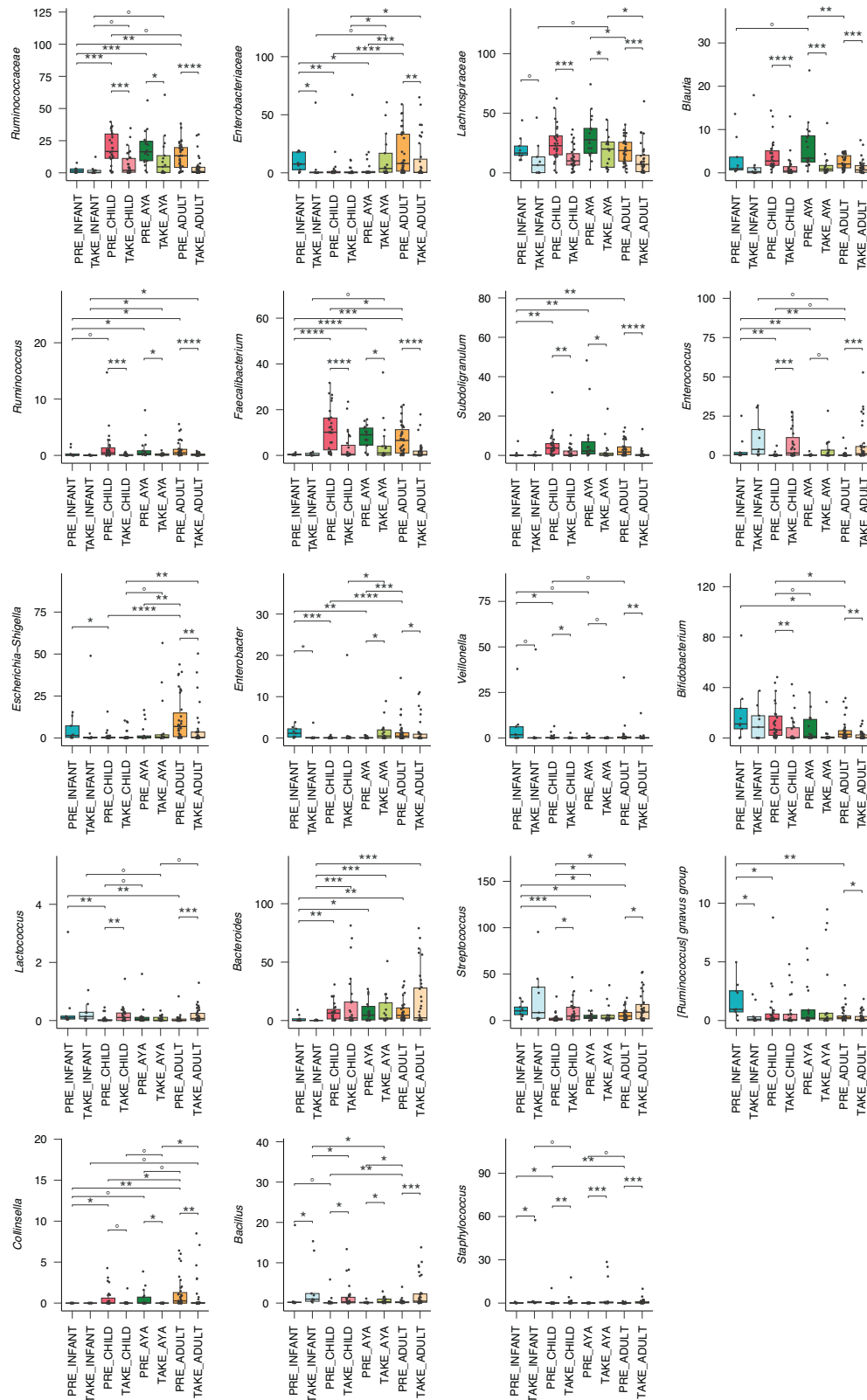


Fig. 4 Gut microbiota genera are differentially represented across ages. Boxplots showing the relative abundance (%) distribution of genera of the gut microbiota differentially represented between samples taken before HSCT ("PRE") and neutrophils engraftment ("TAKE") from patients stratified by age in four groups: 0–3 years ("INFANT"), 4–14 years ("CHILD"), 15–39 years ("AYA") and 40–70 years ("ADULT"). Wilcoxon test, * p value ≤ 0.05 ; ** p value ≤ 0.01 ; *** p value ≤ 0.001 ; **** p value ≤ 0.0001 ; p value ≤ 0.09 .

their more plastic GM [39–41]. Beta diversity analysis revealed significant shifts in overall GM structure post-HSCT in both groups, consistent with earlier studies [19, 37]. Interestingly, while the GM composition of younger and older patients tended to diverge before transplantation, this difference disappeared at neutrophil engraftment. Such a finding suggests that the strong environmental, therapeutic, and immunological pressures associated with HSCT may override intrinsic age-related differences in GM composition. As the age-specific signature is abolished, it converts to a “transplant-specific” signature.

Results at the family and genus levels further highlighted a decline in health-associated and butyrate-producing bacteria (e.g., *Lachnospiraceae*, *Ruminococcaceae*, *Christensenellaceae*, *Bacteroidaceae*, *Blautia*, *Ruminococcus*, and *Faecalibacterium*), with a concomitant rise in potentially pathogenic or opportunistic taxa such as *Staphylococcaceae* (including *Staphylococcus*), *Bacillaceae*, and *Enterococcaceae* (including *Enterococcus*). These compositional changes align with well-established patterns of microbial dysbiosis observed in patients undergoing allo-HSCT [2, 8, 12, 19, 37, 38]. We also found that certain post-transplant GM dynamics differed between age groups: for instance, younger patients uniquely exhibited increases in *Streptococcaceae*, whereas older patients showed a distinctive rise in *Bacteroidaceae* and *Anaerovoracaceae*. Prior to allo-HSCT, older patients displayed higher levels of several taxa (including *Butyricicoccaceae*, *Staphylococcaceae*, *Coriobacteriaceae*, and *Enterobacteriaceae*) compared to younger ones, who had substantially higher levels of *Bifidobacteriaceae* (and *Bifidobacterium*), as described previously [40, 42]. After allo-HSCT, these age-related differences largely vanished, except for a persistent increase detected in adults for certain taxa (e.g., *Enterobacteriaceae* and *Escherichia-Shigella*), reinforcing the idea that the intense selective pressures imposed by allo-HSCT tend to homogenize the microbiome across different ages into the “transplant-specific” signature. K-means clustering corroborated this convergence, revealing distinct GM patterns pre-HSCT (e.g., clusters dominated by *Bifidobacterium* in younger patients), while post-transplant samples from both groups shifted to a more balanced cluster distribution. Interestingly, when analyzing the impact of GM diversity on aGvHD, we confirmed the previously known association and, in a multivariate model, it overwhelmed the age-related signature. Looking at a genus level, we identified a lower abundance of health-associated SCFA-producing genera (e.g., *Eubacterium hallii* group, *Lachnospiraceae* NC2004 group, and other *Lactobacillaceae*), in patients who developed clinically significant aGvHD, pointing to an imbalance in beneficial taxa as a potential risk factor [43, 44]. Conversely, the increase of pro-inflammatory taxa, such as the [*Ruminococcus*] *gnavus* group, was more pronounced in patients with clinically significant aGvHD, especially by neutrophil engraftment [45]. Given the known impact of antibiotics on GM, we accounted for prophylaxis in our analysis [46–48]. While antibiotic exposure was associated with reduced GM diversity, particularly in younger patients, the transplant process itself appeared to play a dominant role in shaping microbiota dynamics. Including antibiotic prophylaxis as a covariate revealed only minor differences in GM composition, suggesting it is not a major confounder of the age- and timepoint-related changes observed. Looking at differences in GM composition of younger and older patients who later develop aGvHD, we again found distinct microbial patterns. Adults exhibited higher levels of certain opportunistic taxa (e.g., *Enterobacteriaceae*, *Escherichia-Shigella*, and *Pseudomonadaceae*), while younger individuals without aGvHD showed an enrichment of beneficial commensals. Notably, specific taxa such as *Bacillaceae* and *Carnobacteriaceae* were more abundant in adults without aGvHD, whereas younger patients who developed aGvHD displayed a depletion of key SCFA-producing bacteria. This may highlight how the depletion of commensal seems to be the main GM signature associated with GvHD in younger patients,

whereas the overgrowth of pathobionts is more typical in older patients.

Stratifying patients into four age-based categories (infants, children, AYA, and adults) provided an even more detailed view of age-related taxa distribution. For instance, *Bifidobacterium* and *Lactococcus* were highest in infants and children before HSCT, which aligns with previous data showing that these genera are key components of the early-life GM [40, 42]. *Veillonella*, also characteristic of younger microbiomes, was highest in infants pre-HSCT and declined in all age groups post-HSCT. In contrast, certain members of the *Enterobacteriaceae* family (e.g., *Escherichia-Shigella* and *Enterobacter*) were more abundant in both infants and adults compared to intermediate age groups before transplant, highlighting potential age-specific colonization patterns that become disrupted after HSCT. Butyrate-producing taxa (*Ruminococcaceae*, *Ruminococcus*, *Faecalibacterium*, and *Subdoligranulum*), known for their beneficial roles in gut barrier function and immune modulation, were generally lower in infants pre-HSCT and declined further post-transplant in all age groups, as previously reported in the HSCT context [8, 38, 49, 50]. Interestingly, the *Lachnospiraceae* family, including *Blautia*, demonstrated similar pre-HSCT patterns and declines across groups post-transplant. In contrast, *Staphylococcus*, an opportunistic bacterium, increased across all age groups, with children showing higher pre-HSCT levels than infants and adults. This may reflect changes in gut permeability and opportunistic expansion post-transplant and may indicate a broader age-independent response to the immunosuppressive effects of HSCT [37]. The number of patients in each of the four age-based categories was not sufficient to perform a clinic sub-analysis. Despite the important insights provided, our study has several limitations. First, the relatively small sample size in both adult and pediatric groups may limit the generalizability of our findings. Second, although the pediatric cohort was recruited from the same hospital and center, their enrollment was asynchronous compared to the adult cohort. This temporal discrepancy could introduce potential confounders, such as evolving clinical protocols and shifts in the hospital environment over time. Third, differences in conditioning regimens between adults and children may have shaped the GM in distinct ways, thereby complicating direct comparisons. Indeed, the presence of a shared antibiotic stewardship program may mitigate these differences [51]. Larger, prospective, and contemporaneous studies that standardize treatment protocols and enrollment timelines are needed to validate and expand on our observations. Future studies should prioritize functional analyses of these microbiome shifts, particularly with regard to immune reconstitution, aGvHD development, and long-term transplant outcomes. Such research may ultimately guide the development of specific microbiome-targeted interventions, ranging from prebiotics and probiotics to fecal microbiota transplantation—that could support beneficial taxa, promote healthier reconstitution of the GM, and potentially improve HSCT outcomes.

In conclusion, our study underscores the broad and uniform impact of HSCT on GM in both adult and pediatric patients. We observed an overall reduction in biodiversity coupled with a shift away from health-associated butyrate producers toward potentially pathogenic taxa. While age significantly influences GM composition prior to transplantation, post-HSCT microbiomes converged across age groups in a “transplant-specific” signature, a pattern that likely reflects the pervasive effects of drugs, allo-immunity, and hospital-based exposures on gut ecology.

DATA AVAILABILITY

The database used for statistical analysis is openly available in Zenodo at 10.5281/zenodo.15262394, under the applicable license. Raw sequencing reads were deposited in the National Center for Biotechnology Information Sequence Read Archive (NCBI SRA; BioProject: PRJNA 1263152).

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AUTHOR CONTRIBUTIONS

DL and MR enrolled and treated patients, designed the study and methodology, collected data, performed formal analysis, data curation and visualization, and wrote the initial draft. SR designed the study and methodology, collected data, performed formal analysis, data curation and visualization, and wrote the initial draft. EM enrolled and treated patients, wrote the original draft of the manuscript. MF designed the study and methodology, collected data, performed formal analysis, data curation and visualization, and reviewed the manuscript. GS, ET, ED, AS, SDM, BS, DM, and NSB collaborated in data curation. AP and FB enrolled and treated patients, designed the study, supervised the data analysis, and reviewed the manuscript. PB designed the study, supervised the data analysis, and reviewed the manuscript. RM enrolled and treated patients, designed the study, supervised the data analysis, reviewed the manuscript, and acquired financial support for the project leading to this publication. All authors read and approved the final manuscript.

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COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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